NAGA KUSHAL VANKADARA

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Education

- **Indian Institute of Information Technology Design and Manufacturing (IIITDM), Kurnool** — *CGPA: 9.11*
  Bachelor of Technology in Computer Science and Engineering — Nov 2022 – May 2026
- **Narayana Junior College, Bachupally, Medchal** — *979/1000 (97.9%)*
  Intermediate (MPC) — Aug 2020 – May 2022

Experience

- **DRDL, Hyderabad** — *Feb 2026 – May 2026*
  AI & Analytics Intern — On-site
  - Engineered a Python-based analytics pipeline to collect and process network metrics (CPU, memory, and traffic) using SNMP, enabling continuous monitoring of device performance.
  - Applied Isolation Forest for unsupervised anomaly detection and implemented **4** specialized detectors for midnight traffic spikes, port flapping, underutilization, and statistical deviations.
  - Designed an interactive React dashboard with real-time visualizations of traffic trends, resource utilization, anomaly scores, and severity distributions, integrating FastAPI, Redis Pub/Sub, and WebSockets to support data-driven monitoring.

Projects

- **Housing Market Analytics**
  Tools: Power BI, Google BigQuery, SQL, DAX, Excel
  - Analyzed **100,000** housing records using Google BigQuery and Power BI to uncover regional pricing patterns, sales trends, and differences across property types.
  - Created **8** DAX measures across **3** interactive dashboard pages to track YoY sales growth, rolling 12-month sales, median price changes, and price per square metre.
  - Designed interactive reports with **4** slicers for area, city, sales type, and region, enabling users to drill into market segments and compare regional performance and property pricing.

- **RetailSphere — Retail Sales Analytics**
  Tools: Databricks, PySpark, SQL, Medallion Architecture, Power BI
  - Built an end-to-end retail analytics pipeline in Databricks using Medallion Architecture to process **1,576** customer, product, and sales records across Bronze, Silver, and Gold layers.
  - Implemented PySpark data cleaning and validation to handle missing values, duplicates, inconsistent formats, and invalid sales references, producing reliable, analytics-ready datasets.
  - Modeled reusable fact and dimension tables and a Power BI dashboard tracking **4** key metrics, including net sales of 1.95M, 1,200 orders, 288 unique customers, and 5,073 units sold.

- **Student Engagement Analytics Automation**
  Tools: n8n, JavaScript, Google Sheets, Gemini AI, HTML/CSS, QuickChart
  - Automated the processing of **1,000** student records using n8n and JavaScript, identifying and resolving duplicate entries, missing values, and inconsistent data formats to ensure standardized data.
  - Defined **6** KPIs and **7** visualizations to analyze student distribution, subscription patterns, academic performance, engagement, retention, and enrollment trends.
  - Integrated Gemini AI to generate actionable business insights and recommendations, delivering an HTML dashboard and automated Gmail reports through an end-to-end workflow.

Skills

- **Programming Languages**: Python, SQL
- **Data Visualization & BI**: Power BI, Tableau, Excel, DAX, Power Query, Matplotlib, Seaborn
- **Data Analysis**: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis
- **Databases & Cloud**: MySQL, MSSQL, Snowflake, Google BigQuery, Azure
- **Data Engineering**: Databricks, PySpark, Delta Lake, ETL Pipelines, Medallion Architecture
- **AI Tools & Automation**: n8n (Workflow Automation), ChatGPT, Claude, Grok, Perplexity, Copilot, Prompt Engineering
- **Tools & Platforms**: Microsoft Fabric, SSMS, Jupyter Notebook, Git, GitHub, VS Code

Research & Publications

- **Magnitude and Phase Aware Spectral Convolutional Neural Networks: Explainable Frequency-Domain Classification** - Research paper accepted at the 11th International Conference on Computer Vision and Image Processing (CVIP 2026), NIT Calicut 2026.

Achievements

- Led a cross-functional team to design and develop an AI-powered Resume Enhancer application. Secured 1st Place among 10+ competing teams at the GDG Hackathon, demonstrating excellence in innovation, technical execution, and real-world impact.
- Served as CSE Branch Secretary for NSS at IIITDM Kurnool and Co-coordinator of the GDG Web Development Team, contributing to student initiatives, team coordination and web development activities.